5.1 Extreme Values of a Function

Name _____ Period ____ Date _____

AP Calculus 5.1 · Student sheet

I can…

- tell absolute extrema from local extrema.
- find the critical numbers of a function.
- find the absolute maximum and minimum of a continuous function on a closed interval.

Vocabulary · fill in the missing word

Absolute maximum — $f(c) \geq f(x)$ for every x in the _____

Local maximum — $f(c) \geq f(x)$ for every x in some open interval _____ c

Critical number — a number c in the _____ where $f'(c) = 0$ or $f'(c)$ does not exist

Extreme Value Theorem — a _____ function on a closed interval $[a, b]$ has an absolute maximum and an absolute minimum

Part 1 · Hills and valleys

$f(x) = x^3 - 3x$ on $[-2, 3]$.

Point	x	f(x)	Slope there
Top of a hill			
Bottom of a valley			
Highest point on $[-2, 3]$			
Lowest point on $[-2, 3]$			

Part 2 · Notes and examples (fill in each note, then work the example)

Extreme Value Theorem: if f is continuous on $[a, b]$, then f has an absolute maximum and an absolute _____ on $[a, b]$.

Example 1

$f(x) = x^2$ on $[-1, 2]$. Find the absolute maximum and minimum values.

Critical numbers: local extrema can only occur where $f'(c) = 0$ or $f'(c)$ _____.

Example 2

Find the critical numbers of $f(x) = x^3 - 6x^2 + 9x + 2$.

Closed interval method: evaluate f at the critical numbers and at the _____; the largest value is the absolute max, the smallest the absolute min.

Example 3

Find the absolute extrema of $f(x) = x^3 - 3x$ on $[-3, 3]$.

Example 4

Find the critical numbers of $f(x) = x^{2/3}$.

Example 5

Find the critical numbers of $f(x) = \sin x + \cos x$ on $(0, 2\pi)$.

Part 3 · Partner practice

Partner 1

Find the critical numbers of $f(x) = x^3 - 12x$.

Partner 2

(AP® Practice) The absolute maximum value of $f(x) = x^3 - 3x^2 + 1$ on $[-1, 3]$ is

(A) 1 (B) 3 (C) −3 (D) 5